

High-Brightness Holographic Projection

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Compiled September 30, 2026

We propose a holographic projection system that achieves high image quality, brightness, and light efficiency. Using a novel light-efficiency loss function, we are able to concentrate more light on the projection region and improve display brightness compared to conventional projectors. Leveraging emerging artificial intelligence-driven computer-generated holography and camera-in-the-loop calibration techniques, we learn a holographic wave propagation model using experimentally captured holographic images and demonstrate state-of-the-art light reallocation performance with high image quality.

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<http://dx.doi.org/10.1364/ao.XX.XXXXXX>

Projection systems should simultaneously realize high image quality, brightness, and illumination utilization. High brightness is essential for creating a comfortable viewing experience under challenging scenarios such as projection under strong ambient light. High illumination utilization reduces power usage and thermal engineering efforts, allowing for more compact and energy-efficient projector architectures.

Such improvements, however, are difficult to be realized using current approaches, which mostly fall under two categories: (i) improving the light engine in traditional displays [1, 2] or (ii) a dual-display configuration [3, 4]. Although the output power and power efficiency of light engines have improved significantly throughout the years, the light utilization of traditional displays, i.e., category (i), is fundamentally limited by amplitude modulation. This process selectively attenuates light using a transmissive or reflective amplitude spatial light modulator (SLM) to generate a desired intensity pattern. This inability to utilize the full output power of the illumination engine leads to excessive power consumption, thermal engineering challenges, and thus low light efficiency and brightness. On the other hand, dual-display configurations, i.e., category (ii), consist of two SLMs, where one phase SLM is used for reallocating light to different parts of the illuminated region, and another amplitude SLM is used for displaying images. Although they are able to achieve high brightness, dual-display architectures are often undesirable due to increased power consumption and cost, bulkier size, and synchronization issues.

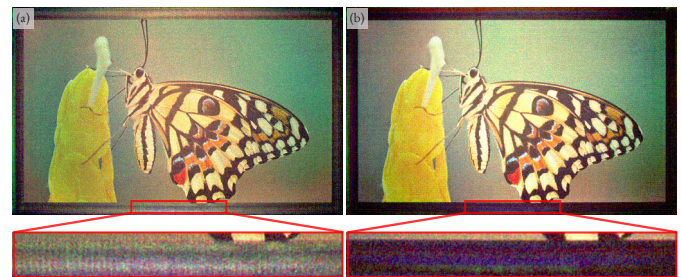


Fig. 1. We propose a holographic projection system that achieves high brightness, light efficiency, and image quality. By optimizing holograms using a camera-calibrated wave propagation model coupled with a new light efficiency loss function, we are able to constrain illumination power to the projection region (center region inside the thick dark border) and thus increase overall brightness, all the while maintaining a high image quality. (a) Image optimized without the loss function. (b) Image optimized with the loss function. The close-ups demonstrate how light wasted outside the image region (left) is reallocated to contribute to the content (right).

Computer-generated holography (CGH), on the other hand, has great potential to revolutionize high-brightness projection systems enabled by its "light reallocation" mechanism where light can in theory be redirected anywhere within the sub-hologram of a point source (please refer to Supplement 1 for more details on the effect of the sub-hologram cone size and light efficiency). This property was experimentally demonstrated in [5] where digital holography is used to perform 3D sensing. However, the aforementioned system as well as prior works on holographic projection systems [6–9] cannot display high-image quality holograms that is absent of speckle noise and ripple artifacts. Furthermore, they did not aim to improve the light efficiency of the holographic displays by improving the CGH optimization process. Recently, emerging artificial intelligence (AI)-driven CGH algorithms [10–13] and camera-in-the-loop (CITL) calibration techniques have achieved state-of-the-art holographic display image quality based on Fresnel holography, yet it is unclear if these frameworks translate well to projectors and they also do not focus on increasing the brightness of the projected images.

Motivated by the aforementioned challenges, we propose a Fresnel holography-based projection system that fully leverages the desirable "light reallocation" property of holography to

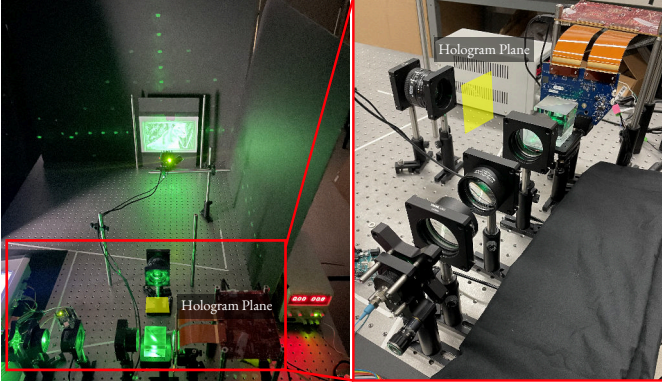


Fig. 2. The holographic projector setup. The holographic image formed on the hologram plane (yellow square) is imaged onto the projection screen.

achieve high-brightness projection. We design a light efficiency-promoting loss function and employ a long propagation distance between phase SLM and the image plane, which improves image brightness and light usage by around 30% while achieving a high image quality. Furthermore, we leverage emerging AI-driven CITL calibration and model training techniques in our projector setup and demonstrate state-of-the-art holographic image quality. Although there is prior work on holographic projection [6–9], their image quality is often unacceptable for a comfortable viewing experience due to the mismatch between physical and simulation setups and their CGH algorithms [14, 15] are not optimized to utilize the full output power of the laser illumination. Our work is the first to leverage AI-driven CGH techniques to drastically improve image quality and increase the brightness of projected holographic images.

Fig. 2 shows our holographic projection system, where the images formed on the hologram image plane (yellow box) via Fresnel holography are projected onto a screen via a projection lens. In Fresnel holography, a collimated coherent laser beam illuminates an SLM with a source field u_{src} . The SLM imparts a phase delay ϕ on u_{src} , resulting in a modulated field u_{SLM} , which is then propagated by a distance z to form a field u_z , and finally imaged onto the projection screen by a magnifying projection lens. The propagated field u_z generates an intensity pattern $|u_z|^2$ on the projection screen due to interference at the hologram plane. The image formation described above is governed by the following equations:

$$\begin{aligned} u_z(x, y, \lambda) &= f(u_{\text{SLM}}(x, y, \lambda), z), \\ u_{\text{SLM}}(x, y, \lambda) &= e^{i\phi(x, y, \lambda)} u_{\text{src}}(x, y, \lambda), \end{aligned} \quad (1)$$

where λ is the wavelength of the laser, x, y are the spatial coordinates, and f is the composition of wave propagation from the SLM plane to the hologram plane and the projection onto the screen. The wave propagation component in f can either be the angular spectrum method [16] or a camera-calibrated learned propagation wave model, and we denote the corresponding f by f_{ASM} and f_{model} , respectively. The latter is described in detail in recent works [10–12] on AI-driven holographic near-eye displays. We employ this learned wave propagation framework in our high-brightness projection system. Here, we use the same model design proposed by Choi et al. [11]. Details on the model architecture are described more in Supplement 1.

Similar to Choi et al. [11], we capture a train and a test set

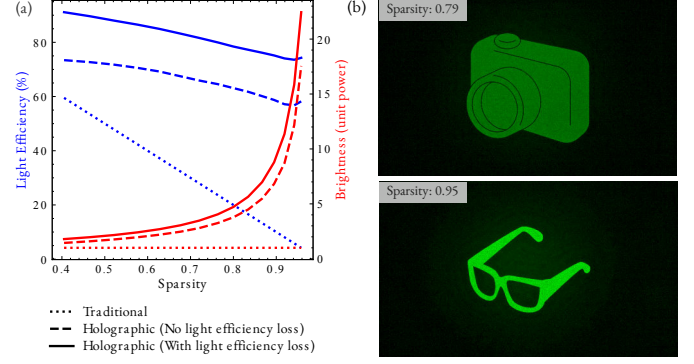


Fig. 3. (a) Quantitative evaluation of image brightness and light efficiency with respect to image sparsity in simulation. For traditional light-attenuating displays, the brightness stays constant as image sparsity increases. On the contrary, there is a hyperbolic increase in image brightness for holographic displays, which is also demonstrated in the experimentally captured images in (b). Although both holographic and conventional displays experience a drop in light efficiency as sparsity increases, holographic displays retain an efficiency of over 70% while the efficiency of conventional displays goes to nearly 0% for extremely sparse content. Our proposed loss function further improves light efficiency and brightness. Here, light efficiency is defined by the ratio between the total power of the light leaving the display and the total power of the light output by the illumination engine, such as a laser or LCD backlight.

comprised of a large number of SLM phase patterns and corresponding amplitude images recorded at a fixed distance. Using a standard stochastic gradient-descent solver, we fit the model parameters on the training image pairs to learn the calibrated holographic projection model.

After the holographic projection model is learned, we can then optimize for the phase pattern ϕ displayed on the SLM that generates a desired target image amplitude a_{target} by minimizing $\mathcal{L}_{\text{recon}}(s \cdot |f_{\text{model}}(e^{i\phi}, z)|, a_{\text{target}})$, where s is a scale factor that is optimized along with ϕ and $\mathcal{L}$ is an arbitrary loss function between the reconstructed image and the target image. We use a stochastic gradient descent (SGD) procedure to optimize for ϕ , and the update rules of our SGD-based solver in iteration k with learning rate α are

$$\phi^{(k)} = \phi^{(k-1)} - \alpha \left(\frac{\partial \mathcal{L}}{\partial \phi} \right)^T \mathcal{L}(s \cdot |f_{\text{model}}(u_{\text{SLM}}, z)|, a_{\text{target}}). \quad (2)$$

As demonstrated in Fig. 1, good image quality can be achieved using a simple image reconstruction loss for $\mathcal{L}$, such as an L_2 loss, yet much light is distributed outside of the region of interest, limiting the light efficiency of the projector and brightness of the projected images. Here, we propose a light efficiency loss function $\mathcal{L}_{\text{reg}}$ that enforces light to be constrained within the region of interest to improve the brightness of the projected images. $\mathcal{L}_{\text{reg}}$ enforces the complex field amplitude to exactly match a_{target} by a scale factor:

$$\mathcal{L}_{\text{reg}} = L_2 \left(|f(u_{\text{SLM}}, z)|, \sqrt{\frac{\sum |u_{\text{src}}|^2}{\sum a_{\text{target}}^2}} a_{\text{target}} \right) \quad (3)$$

This enforces that all illumination power $\sum |u_{\text{src}}|^2$ be distributed within the region of interest while maintaining accurate pixel-

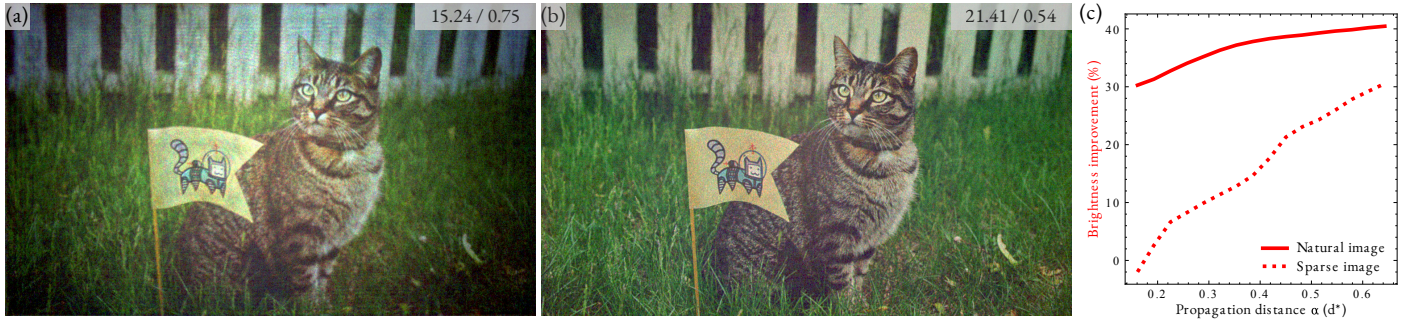


Fig. 4. (a) Experimentally captured projection images optimized using ASM propagation. Severe fringe and chromatic artifacts are visible due to ASM incorrectly modeling long-distance propagation. We also observe a brightness falloff in the peripheral regions due to laser illumination non-uniformity. We report the PSNR/LPIPS of reconstructed images. (b) Experimentally captured projection images optimized using a learned wave propagation model. The model learns to correctly compensate for the long propagation distance artifacts and the illumination non-uniformity. (c) Quantitative evaluation of the light efficiency loss function performance with respect to propagation distance in simulation. As propagation increases, the light efficiency loss function becomes more effective at constraining all the light to the region of interest, improving brightness and light efficiency. This effect is even more significant for sparse images, as the loss function performance improves from nearly 0% (no light reallocation) to 30%.

wise image reconstruction. In practice, $|u_{\text{src}}|^2$ needs to be precisely calibrated and we detail this procedure in Supplement 1.

The total loss of the phase optimization procedure is therefore

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \gamma \mathcal{L}_{\text{reg}} \quad (4)$$

where γ determines the weighting between image reconstruction and light efficiency loss function. Details on the optimization procedure and choosing the appropriate regularization weight are described more in Supplement 1.

Our holographic projector uses a TI DLP6750Q1EVM SLM with a resolution of 1280×800 , a pixel pitch of $10.8 \mu\text{m}$, and a bit depth of 4 bits per pixel. The laser is an FISBA RGBeam fiber-coupled module with three optically aligned laser diodes with a maximum output power of 50 mW and RGB wavelengths of 636.4, 517.7, and 440.8 nm, respectively. A Nikon AF Nikkor 50mm f/1.4D lens is used to magnify the holographic image and project it on a screen 87 cm away from the lens. With this configuration, our projection system provides a large viewing region of red $12 \text{ cm} \times 21 \text{ cm}$. The SLM phases are optimized separately for each color channel and the captured monochrome images are combined in post-processing to create a full-color image. All holograms are eight-frames time-multiplexed, i.e. eight phase patterns are displayed sequentially to generate a single amplitude image. All images are captured with a FLIR Grasshopper 2.3 MP color USB3 vision sensor. Additional hardware and software implementation details are discussed in Supplement 1.

Probably the most desirable feature of holographic projectors is their ability to redirect illumination power to arbitrary regions within a sub-hologram cone. This property enables high-brightness projection through light reallocation by steering light away from dim image regions to bright image regions without blocking any light. Thus, holography can potentially outperform traditional light-attenuating transmissive displays or self-emissive displays in output illumination power utilization and brightness. This is true especially when displaying extremely sparse content where most pixel values are zero, since most light would then be blocked in light-attenuating displays and self-emissive pixels cannot output their maximum power. This can be seen in Fig. 3 – image brightness drastically increases as sparsity increases for holographic projectors but stays con-

stant in traditional displays. Our proposed loss function further improves the brightness at every sparsity level, highlighting our method's robustness to sparsity. Here, we define sparsity as the ratio of zero-valued pixels and the total number of pixels in an image. These results suggest that holographic projection could be extremely useful in applications such as automotive head-up display (HUD) devices, where most of the displayed content is sparse text and pictograms.

To fully unlock the light reallocation capability of holography, the sub-hologram cone of each SLM pixel should ideally fully cover the projection region such that light can be redirected anywhere within. In other words, the holographic image should be formed very far away from the SLM, which means that the propagation distance z should be long ($z \geq 0.6 d^*$). More discussion regarding the normalized distance unit d^* can be found in Supplement 1. This is illustrated in Fig. 4(c), where we show that a long propagation distance is crucial for a holographic display to perform light reallocation. However, we observe from Fig. 4(a) that a long propagation distance results in severe fringe artifacts in the captured SGD-optimized images. This is the reason why prior most works employ a near-SLM setting that performs little to no light reallocation [13, 17]. In contrast, the learned wave propagation model in our work fully enables a long propagation distance, as shown in Fig. 4(b) where all artifacts are eliminated. Coupled with our new loss function, we are able to optimize light efficiency and image brightness while maintaining great image quality.

From Figs. 1 and 5, we observe that the light efficiency loss function successfully constrains light within the projection region and eliminates all stray light outside of the region, effectively improving image brightness by around 30% for all test cases. Furthermore, the pixel-wise and perceptual image reconstruction performance is also maintained when images are optimized with the loss function, showing that a significant amount of light efficiency can be achieved without a large drop in image quality.

To summarize, we demonstrate that our holographic projection system is capable of displaying images with unprecedented image quality, brightness, and light efficiency. Nevertheless, our method is not without limitations. It takes about 20 minutes to optimize a phase pattern on a single NVIDIA RTX A6000.

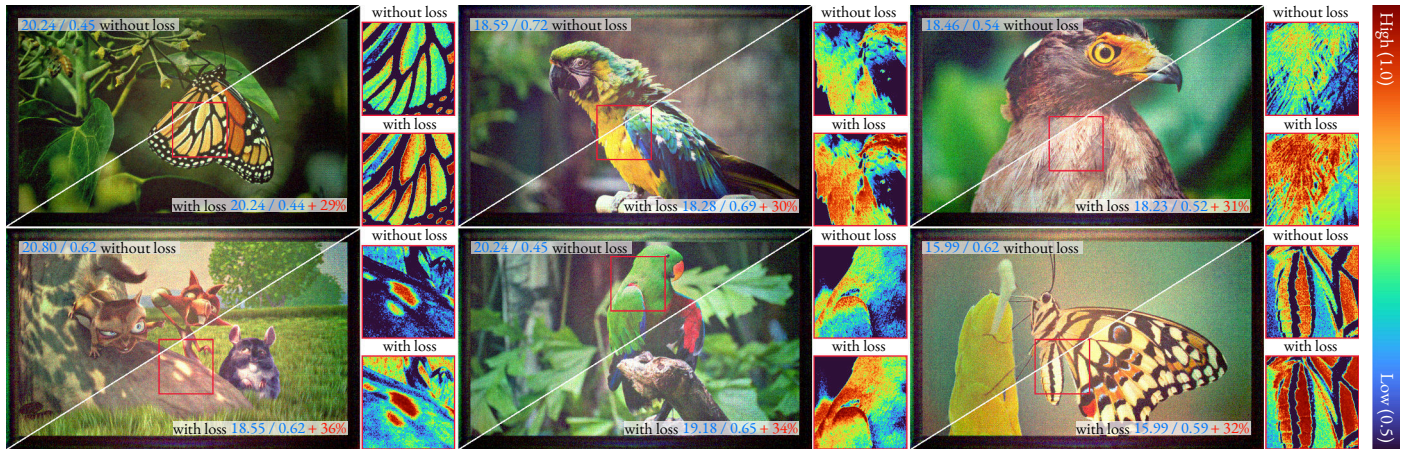


Fig. 5. Comparison of experimentally captured images optimized without and with the light efficiency loss function. We report the image reconstruction quality (PSNR/LPIPS) in blue text and brightness improvement in red. We also show color-mapped zoom-in crops of the luminance channel, where pixel values below 0.5 are clipped to 0.5 and the rest linearly scaled to [0.5, 1]. By optimizing holograms using our proposed loss function, stray light can be completely eliminated outside of the region of interest, effectively increasing the overall image brightness by around 30%. The LPIPS performance for images optimized with the loss function is retained, indicating that there is no significant drop in perceptual image quality. See Visualization 1 for a split-screen animation comparison.

Although it is possible to train an inverse model that directly maps input images to optimized phases [12, 13], we have not attempted to do these in this letter and we leave it for future work. Currently, the zeroth order undiffracted light in our system adversely affects the dynamic range of the projected content and reduces image contrast. However, we believe that this challenge can eventually be overcome by using higher-efficiency SLMs or modifying the system to an off-axis setup where the images and the zeroth order DC term could be separated by applying a phase ramp to the SLM [18, 19]. Finally, we have not extended this method to video content, but in theory it can be easily implemented by dynamically adjusting the laser power to account for the different scale factor s values across frames.

The powerful light reallocation mechanism of holography together with AI-driven, fully automatic display calibration techniques has great potential to revolutionize high-brightness projection systems. In this work, we leverage state-of-the-art camera-calibrated wave propagation model algorithms and a novel loss function design to achieve superior image quality, brightness, and light efficiency in a holographic projection system.

Funding. This project has no associated award funding.

Acknowledgments. B.C. is supported by the Stanford Graduate Fellowship (SGF) and the NSF Graduate Research Fellowship. M.G. is supported by the SGF. S.C. is supported by the Meta Research Ph.D. Fellowship and the Kwanjeong Scholarship. We thank Sony and Samsung for helpful discussions and support.

Disclosures. The authors declare no conflicts of interest.

Data availability. All data needed to evaluate the conclusions in the Letter are present in the Letter and/or Supplementary Materials.

Supplemental document. See Supplement 1 for supporting content.

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Supplement High-Brightness Holographic Projection

1. HARDWARE IMPLEMENTATION

A. Details on Benchtop Setup

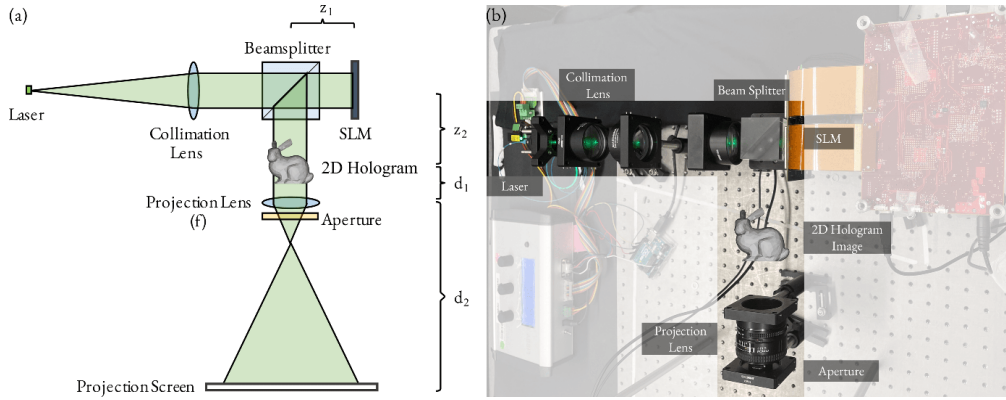


Fig. S1. (a) Schematic diagram of the proposed holographic projector. (b) Physical setup of the holographic projector. The light emitted by the single-mode laser fiber is first collimated by a 100mm lens before arriving at the beam splitter, and then reflected off the SLM. The SLM-modulated field forms a 2D Fresnel hologram, which is then relayed onto the projection screen by a projection lens. An additional aperture in front of the projection lens is used to filter out the higher diffraction orders.

In addition to the hardware details listed in the main paper, we show additional photographs of our setup in Fig. S1. The distance values in in Fig. S1(a) satisfy the following relationships:

$$\begin{aligned} z_1 + z_2 &= z \\ \frac{1}{d_1} + \frac{1}{d_2} &= \frac{1}{f} \\ z_2 + d_1 &= C \end{aligned} \tag{S1}$$

where z is the propagation distance from the SLM to the 2D hologram, z_1 is the distance from the SLM to the beam splitter, z_2 is the distance from the beam splitter to the 2D hologram, f is the focal length of the projection lens, d_1 is the distance from the 2D hologram to the projection lens, and d_2 is the distance from the projection lens to the projection screen.

In our setup, the propagation distance z is around 20 cm. A Nikon AF Nikkor 50mm f/1.4D lens is used to magnify the holographic image and project it on a screen $d_2 = 87$ cm away from the lens. From S1, we can compute that $d_1 = 5.3$ cm. Hence, the magnification of the projection system is $M = \frac{d_2}{d_1} = \frac{87}{5.3} = 16.4$.

We use a TI DLP6750Q1EVM SLM133 with a pixel pitch of $10.8 \mu\text{m}$. The region of interest (ROI) on the SLM is 700×1190 pixels, which translates to an area of $7.56 \times 12.85 \text{ mm}^2$. Hence, the projection area is $16.4 \cdot 7.56 \times 16.4 \cdot 12.85 \text{ mm}^2 = 12.4 \times 21.1 \text{ cm}^2$.

B. Propagation Distance Calibration

To generate high-quality holograms, our method requires precise calibration of the distance z , which is the propagation distance from the SLM to the 2D hologram plane. Note that once the

setup is fixed, the distances z_1 , d_2 , and $z_2 + d_1$ are all constant, which are expressed as boldface in Eq. (S1). Hence, the propagation distance z can be directly expressed as

$$z_1 + C - \frac{d_2 f}{f + d_2} \quad (\text{S2})$$

In practice, we first obtain a coarse estimate of the propagation distance $\hat{z}$ we hope to work with in simulation, which we will describe in Section 2. Next, we implement the hardware such that Eq. (S2) roughly matches the coarse propagation distance. We do a per-color channel calibration step where we run camera-in-the-loop [1] optimization on a test image using different propagation distances in the range $(\hat{z} - \Delta z, \hat{z} + \Delta z)$. We record the distance z^* that corresponds to the best image reconstruction performance, which serves as the final estimate of the propagation distance.

C. Planar Homography Calibration

In the homography calibration step, we use a similar procedure described in [1] and apply a planar homography to accurately register the captured projected images to the ground truth target images. Our implementation uses a target binary pattern consisting of 18×11 white dots where every dot is spaced 70 pixels apart. Therefore, the region of interest (ROI) has 700×1190 pixels and the projection area is around $12.4 \times 21.7 \text{ cm}^2$.

D. Laser Power Calibration

In order to incorporate the light efficiency loss function in our optimization process, it is important that we know the maximum intensity sum $\sum |u_{\text{src}}|^2$ that could possibly be recorded by the camera sensor. To measure this, we display a zero-phase pattern on the SLM and capture the output intensity on the display screen. A constant zero-phase pattern indicates no phase modulation of the collimated laser beam at the SLM plane, and thus the reflected wavefront should simply be a plane wave. We perform this calibration to compute $\sum |u_{\text{src}}|^2$ for all three color channels.

2. THE EFFECT OF PROPAGATION DISTANCE ON LIGHT EFFICIENCY AND IMAGE BRIGHTNESS

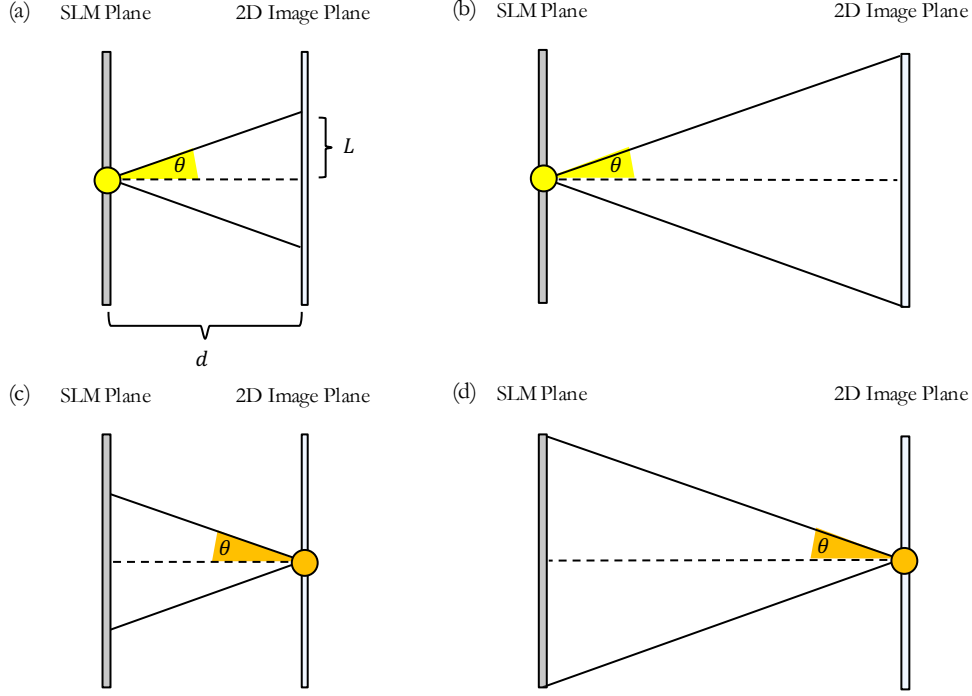


Fig. S2. (a) The sub-hologram cone of an SLM pixel. The propagation d is proportional to the radius L of the sub-hologram cone. (b) When the sub-hologram of an SLM pixel covers the whole projection region, the light modulated by the SLM pixel can in theory be redirected anywhere within the projection region. (c) We can also define the sub-hologram cone of a pixel on the image plane, which is essentially the flipped version of (a). From this perspective, a pixel in the image *receives* light from all SLM pixels within the sub-hologram cone. (d) When the sub-hologram cone of an image pixel covers the whole SLM region, all pixels from the SLM contribute to the image formation of that pixel.

We denote the pixel pitch of the SLM by p , the spatial resolution of the SLM by $H \times W$, the maximum diffraction angle of an SLM pixel by θ , the laser wavelength by λ , and the propagation distance from the SLM plane to the image plane by d . The maximum diffraction angle θ of an SLM pixel can be computed by $\theta = \sin^{-1} \frac{\lambda}{2p}$.

In Fresnel holography, the light diffracted by an SLM pixel can in theory reach anywhere within a disk of radius L on the 2D image plane. The extent to which the diffracted light from a single SLM pixel can reach is often referred to as the sub-hologram cone [2, 3]. The linear relationship between L and d is illustrated in Fig. S2(a) and Eq. S3.

$$\frac{L}{d} = \tan \theta \implies \frac{L}{d} = \tan \sin^{-1} \frac{\lambda}{2p} \implies d = \frac{L}{\tan \sin^{-1} \frac{\lambda}{2p}} \quad (\text{S3})$$

Conversely, light from all SLM pixels within the disk of radius L can reach a single point on the image plane. Hence, to achieve high brightness and light efficiency projection, it is essential that we implement our system with a large sub-hologram cone, and incidentally, a long propagation distance d .

As shown in S3, there propagation distance d depends on the wavelength λ of the laser and the pixel pitch p of the SLM. Furthermore, to reconstruct a hologram with a larger spatial extent (i.e. larger SLM), L also needs to increase. To characterize the propagation distance in a display-agnostic fashion, it makes sense to express the d in terms of normalized unit d^* instead of metric length. Here, we define the value d^* by

$$d^* = \frac{R}{\tan \sin^{-1} \frac{\lambda}{2p}} = \frac{\sqrt{H^2 + W^2}p}{2 \tan \sin^{-1} \frac{\lambda}{2p}} \quad (\text{S4})$$

where $R = \frac{\sqrt{H^2 + W^2}p}{2}$ is the radius of the smallest circle that covers the full SLM ROI. Now, d can be expressed as $d = \alpha d^*$ and the dimensionless value α can be used to characterize propagation distance in a display-agnostic fashion. For instance, for displays with $\alpha = 1$, the subhologram cone of the center pixel of the SLM covers the whole imaging region on the 2D image plane, as shown in Fig. S3. Given a display, larger α also corresponds to a larger propagation distance as the relationship between α and d is linear.

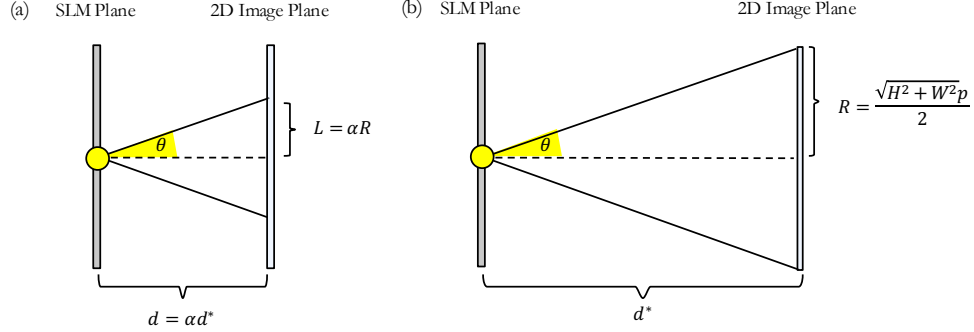


Fig. S3. As shown in the diagram, d^* is determined by the fixed parameters given a display architecture and can be used as a normalized unit. The dimensionless value α is then used to express the propagation distance.

In our system, the pixel pitch $p = 10.8 \mu\text{m}$. The size of the SLM ROI is $7.56 \times 12.85 \text{ mm}^2$, hence $R = \frac{\sqrt{7.56^2 + 12.85^2}}{2} = 7.45 \text{ mm}$. We let λ be the wavelength of the green laser, which is 517 nm . Hence, $d^* = 31.11 \text{ cm}$.

From simulation results in Fig. S4, we see that the light efficiency of the projection system peaks around $20 \text{ cm} = 0.643 d^*$. The final calibrated propagation distances for red, green, and blue channels are 19.7 , 19.9 , and 20.1 cm , respectively, and corresponds to α values of 0.633 , 0.639 , 0.643 .

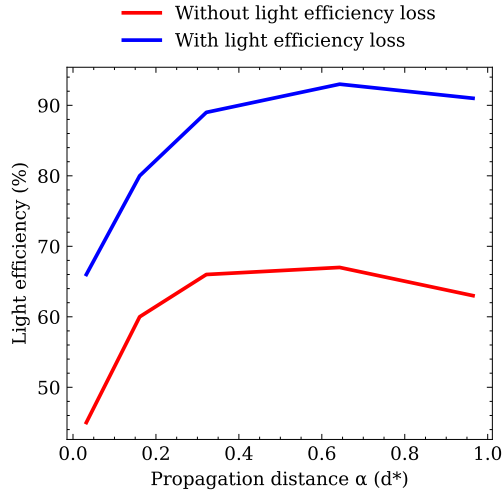


Fig. S4. Light efficiency with respect to propagation distance in simulation. The light efficiency performance is evaluated on the whole testing dataset. We see that the light efficiency peaks when the propagation distance is around $0.64 d^*$ (20 cm)

3. SOFTWARE IMPLEMENTATION

We implement all optimization algorithms and model training procedures in PyTorch, and we describe our implementation in detail in the following sections.

A. Iterative Phase Optimization

The pseudocode for both SGD-ASM and SGD-model is outlined in Alg. S1. The propagation operator f_p could either be the angular spectrum method f_{ASM} [4] or a learned propagation model f_{model} [1, 5], which corresponds to the SGD-ASM and SGD-model algorithm, respectively.

Other than SGD using a learned model and ASM, we also perform SGD with camera-in-the-loop (CITL) optimization [1]. Details of the CITL process are explained thoroughly in various prior works [1, 6, 7], and we outline the pseudocode in Alg. S2.

We initialize the phase pattern with values in the range $(-0.5, 0.5)$ and initialize s to 1. The scaling factor s is updated in each iteration to minimize the mean-squared-error between $s \cdot a_{\text{recon}}$ and a_{target} :

$$s^* = \underset{s}{\operatorname{argmin}} \|s \cdot a_{\text{recon}} - a_{\text{target}}\|^2 = \frac{a_{\text{recon}}^T a_{\text{target}}}{a_{\text{target}}^T a_{\text{target}}} \quad (\text{S5})$$

where a_{recon} is the amplitude of the predicted 2D hologram image and a_{target} is the amplitude of the target image, both in vectorized form. We run the phase optimization for 2000 iterations with a learning rate of 0.02. To optimize heavily quantized phase-only SLMs, we follow the same optimization protocol using Gumbel-Softmax as described in Choi et al. [5] to deal with discrete (in our case 4 bits, 16 levels) phase levels.

Algorithm S1. SGD Hologram Optimization

$\text{f_prop}(\cdot, z)$: propagate a field through any propagation model f_p for a distance z
 $\text{f_backprop}(\cdot)$: backpropagation through f_p
 $\text{loss_backprop}(\cdot)$: backpropagation through loss function
 s : scaling factor.
 a_{target} : amplitude of target image
1: **foreach** k in $1 \dots K$ **do**
2: $\hat{u} \leftarrow \text{f_prop}(e^{i\phi}, z)$
3: $s \leftarrow \frac{|\hat{u}|^T a_{\text{target}}}{a_{\text{target}}^2} \quad \triangleright \text{Compute scaling factor to minimize } \|s \cdot |\hat{u}| - a_{\text{target}}\|^2$
4: $\phi \leftarrow \text{f_backprop}(\text{loss_backprop}(\mathcal{L}(s \cdot |\hat{u}|, a_{\text{target}})))$
5: **return** ϕ

Algorithm S2. SGD-CITL Hologram Optimization

$\text{f_prop}(\cdot, z)$: propagate a field through any propagation model f_p for a distance z
 $\text{camera_prop}(\cdot)$: camera captured and homography-corrected intensity (raw mode)
 $\text{f_backprop}(\cdot)$: backpropagation through f_p
 $\text{loss_backprop}(\cdot)$: backpropagation through loss function
 $\text{replace}(s, t)$: replace values of s with t , but retain gradients from s
 s : scaling factor
 a_{target} : amplitude of target image
1: **foreach** k in $1 \dots K$ **do**
2: $\hat{u} \leftarrow \text{f_prop}(e^{i\phi}, z)$
3: $I \leftarrow \text{camera_prop}(\phi) \quad \triangleright \text{Display phase pattern } \phi \text{ on SLM and capture}$
4: $s \leftarrow \frac{|\hat{u}|^T a_{\text{target}}}{a_{\text{target}}^2}$
5: $\phi \leftarrow \text{f_backprop}(\text{loss_backprop}(\mathcal{L}(s \cdot \text{replace}(|\hat{u}|, \sqrt{I}), a_{\text{target}})))$
6: **return** ϕ

B. Model Architecture and Training Details

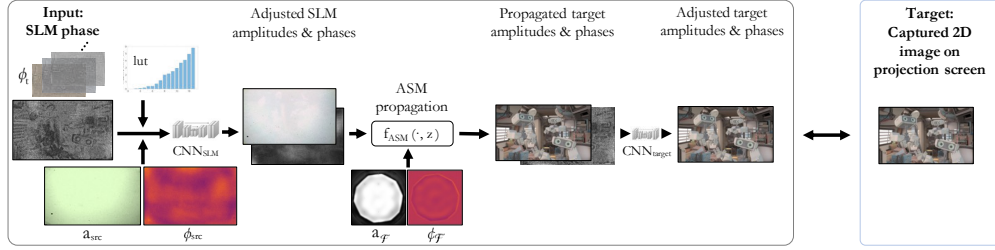


Fig. S5. Network architecture of our calibrated wave propagation model.

For the learned wave propagation model, we use the same architecture as in Choi et al. [5]. Since we are only working with 2D projection, we train our model on one fixed propagation z , which is the distance from the SLM to the 2D hologram plane. As shown in [6], the distance from the SLM to the hologram plane needs to be calibrated per color channel, and the final distance measurements are 19.7, 19.9, and 20.1 cm for red, green, and blue channels, respectively. We use a batch size of 4 and a learning rate of $4e^{-4}$. We generate 4800 phases using SGD-ASM for each color channel and captured the intensities on the projection screen to generate a paired dataset. Instead of capturing images formed on the hologram plane, we capture the images displayed on the projection screen since this would allow the model to additionally learn to compensate for the existing artifacts induced by the projection screen. The dataset is partitioned into train, validation, and training sets with a ratio of 8:1:1. The training takes around 24 hours to converge on an NVIDIA A6000 GPU.

4. EVALUATION OF BRIGHTNESS

A. Characterizing Brightness

Recall that the scaling factor s is computed at each update step to minimize the mean-squared-error between $s \cdot a_{\text{recon}}$ and a_{target} . Intuitively, the factor s can be interpreted as the amplitude of the laser illumination that is needed such that the captured/reconstructed image intensity best matches the target image intensity. Hence, a small s corresponds to lower laser power and high image brightness, and vice versa.

Suppose the scaling factor s of the three color channels is defined by s_R, s_G, s_B . The total power budget to display a full-color hologram is therefore $s_R^2 + s_G^2 + s_B^2$. The brightness of the full-color hologram can therefore be characterized as $\frac{1}{s_R^2 + s_G^2 + s_B^2}$.

B. Comparison with Conventional Displays

The peak brightness of conventional amplitude modulation displays is limited by the maximum output power of the pixels in the backlight panel L . When displaying a target image I using conventional displays, the measured intensity of the displayed image would be $I \cdot \frac{\max(L)}{\max(I)}$. The scaling factor s is therefore $\sqrt{\frac{\max(I)}{\max(L)}}$. Similar to the calculation in Section A, the brightness of a full-color image displayed using conventional displays can be characterized as

$$\frac{1}{\frac{\max(I_R)}{\max(L_R)} + \frac{\max(I_G)}{\max(L_G)} + \frac{\max(I_B)}{\max(L_B)}} \quad (\text{S6})$$

where I_R, I_G, I_B are the RGB channels of the image, and L_R, L_G, L_B are the maximum output power of the RGB pixels.

To mimic conventional displays using our hologram display engine, we assume that our reflective phase SLM operates purely as a mirror and does not perform phase modulation. Each SLM pixel is then analogous to the backlight pixels of conventional displays. The maximum output power of each pixel is thus $\max(L) = \frac{P}{H \times W}$, where P is the calibrated total laser output power (See 1-C) and H, W are the spatial resolutions of the SLM. From Fig. S6, we see that under a fixed power budget, conventional displays are much dimmer compared to holographic displays.

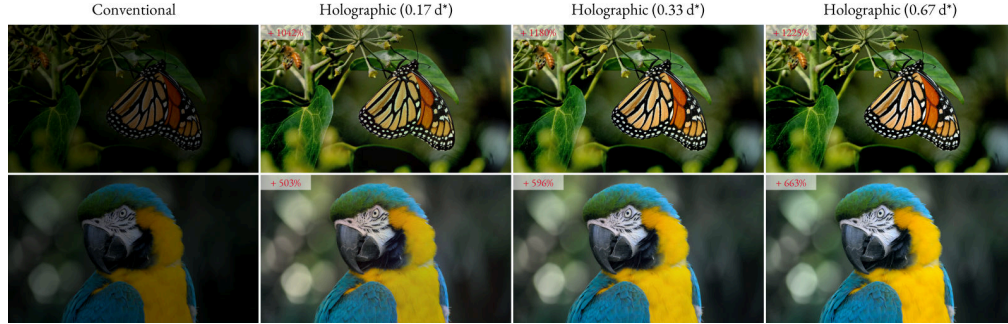


Fig. S6. Brightness of holographic displays compared with conventional amplitude modulation displays under a fixed power budget in simulation. We see that holographic displays can achieve much higher brightness given a fixed power budget. Furthermore, the brightness increases as the propagation distance in the holographic display increases.

5. THE EFFECT OF PROPAGATION DISTANCE ON IMAGE QUALITY

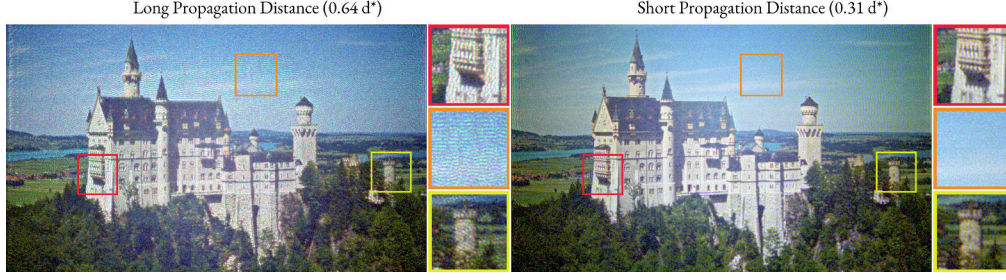


Fig. S7. Comparison of experimentally captured images optimized using different propagation distances. Visible wave-like artifacts are shown in the zoom-in crops of the long propagation distance ($z = 0.64 d^*$) case.

To perform light reallocation effectively, the propagation distance z from the SLM plane to the 2D hologram plane needs to be long enough such that the sub-hologram cones of the SLM pixels cover the whole projection region. However, it is generally hard to produce good image quality using a long propagation distance due to tricky issues in frequency domain sampling. There has been extensive research on more clever wave propagation algorithms that improve holographic image quality [8, 9].

Here, we show SGD with camera-in-the-loop (CITL) [1] optimization results using different propagation distances. As shown in Fig. S7, long propagation distances give rise to severe fringe artifacts and chromatic blur, suggesting that naive SGD might not work with long propagation distances. Hence, we leverage a learned wave propagation model to compensate for these artifacts.

6. THE EFFECT OF LIGHT EFFICIENCY LOSS FUNCTION WEIGHT



Fig. S8. Qualitative and quantitative results of varying the light efficiency loss function regularization weight. By increasing the regularization weight, we are able to constrain more light to the projection region.

We show experimentally captured 2D hologram results optimized using different light efficiency loss function weights in Fig. S8. As the loss function weight increases, we observe that more light is restricted within the projection region, improving image brightness. However, the image quality slightly drops. Thus, the light efficiency loss function weight is a parameter that should

be tuned depending on the application. For example, a large weight should be used if display brightness is prioritized over image quality, such as outdoor projection under sunlight.

7. EXPERIMENTS ON ASM PROPAGATION AND CAMERA-IN-THE-LOOP (CITL) OPTIMIZATION

Other than SGD-model presented in the main paper, we also show images optimized using an ideal wave propagation model (SGD-ASM) and camera-in-the-loop (CITL) techniques [1] (SGD-CITL) in Fig. S9 and compare with SGD-model. For the ideal wave propagator, we use the angular spectrum method [4]. Our light efficiency loss function works with both cases although less significant, demonstrating the importance of using a camera-calibrated learned wave propagation model. Similar to SGD-model, SGD-CITL corrects the illumination nonuniformity artifacts present in SGD-ASM but generates fringe artifacts. We suspect that this is mainly due to the incorrect gradients that arise from long propagation distances accumulating over CITL optimization steps.

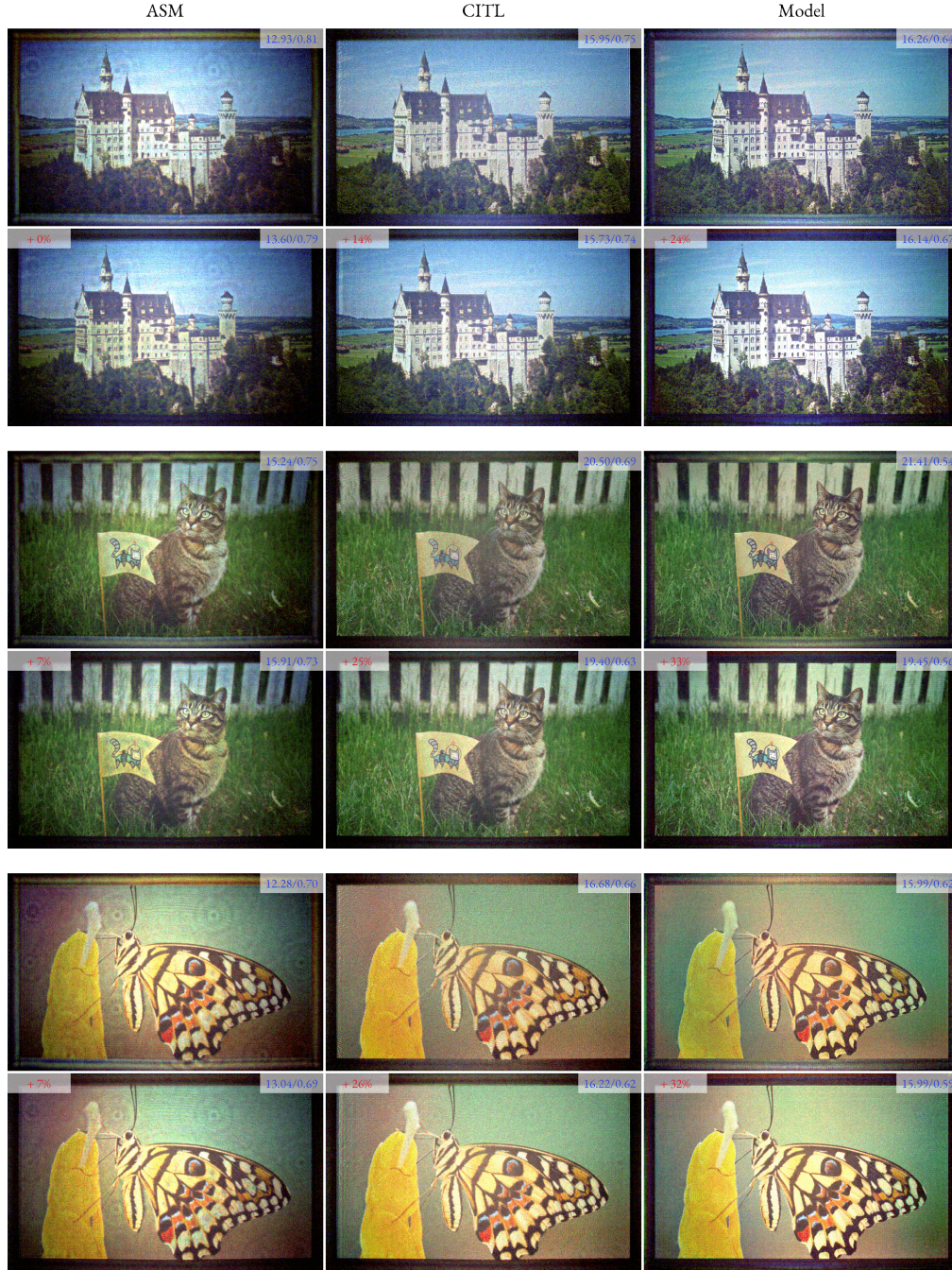


Fig. S9. Comparison of experimentally captured images optimized using different methods. The light efficiency loss function also works well for SGD-CITL, but wave-like artifacts arise in the final optimized image. SGD-model generates clean results with no noticeable artifacts and has the best brightness improvement when optimized with the light efficiency loss function.

8. ADDITIONAL RESULTS ON SGD-MODEL

This section includes additional captured results in support of those shown in the paper for different target images. The light efficiency loss function successfully improves the brightness and retains image quality in all cases. We also show a sparse image test result, where the brightness improvement is less significant.

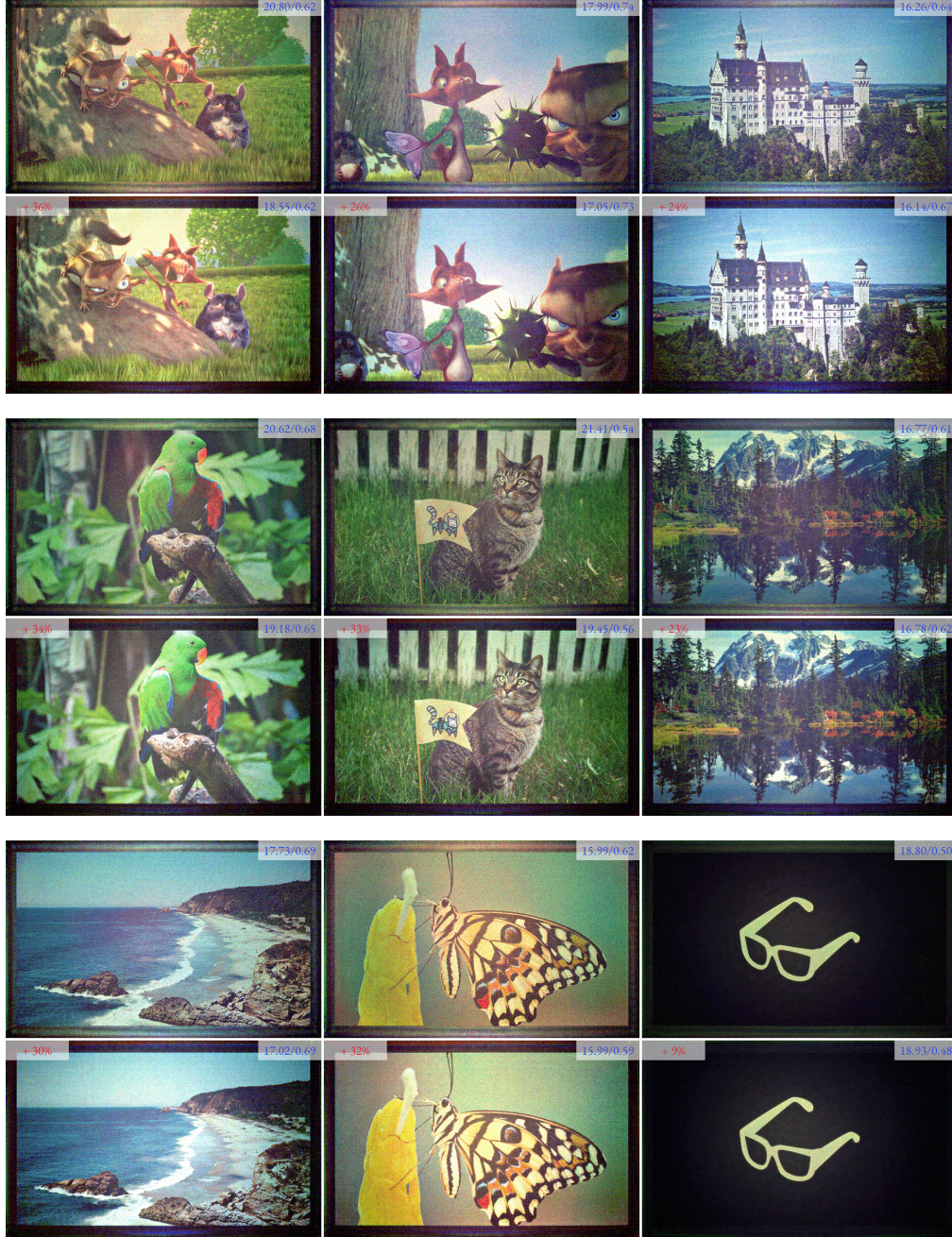


Fig. S10. The light efficiency loss function improves the brightness of projected images by around 20-30% for all natural images while maintaining image quality.

9. THE EFFECT OF ROI SIZE ON IMAGE BRIGHTNESS AND IMAGE RECONSTRUCTION QUALITY

In our main setup, the native resolution of the SLM is 800×1280 pixels and the resolution of the region of interest (ROI) on the SLM is 700×1190 pixels. This translates to a large projection region of $12 \times 21 \text{ cm}^2$ after the 2D hologram is magnified onto the projection screen. SLM pixels outside of the ROI are not optimized.

The reason we used a ROI slightly smaller than the SLM resolution is illustrated in Fig. S11. It is impractical to set the ROI to the full support of the wavefield or a spatial extent much larger than the SLM area because such a setting would lead to bad image quality, as shown in Fig. S11. The degrees of freedom of the SLM pixels are not enough to generate high-image quality holograms with spatial resolution significantly larger than the native SLM resolution. As the ROI decreases, we see that the image reconstruction performance gradually improves. This is why most prior works in CGH [1, 2, 5–7] have an ROI slightly smaller than or equal to the native SLM resolution. Our goal is to improve the brightness of the system while maintaining image quality for a comfortable viewing experience. Hence, we follow prior works and also employ the setting where the ROI is slightly smaller than the SLM resolution.

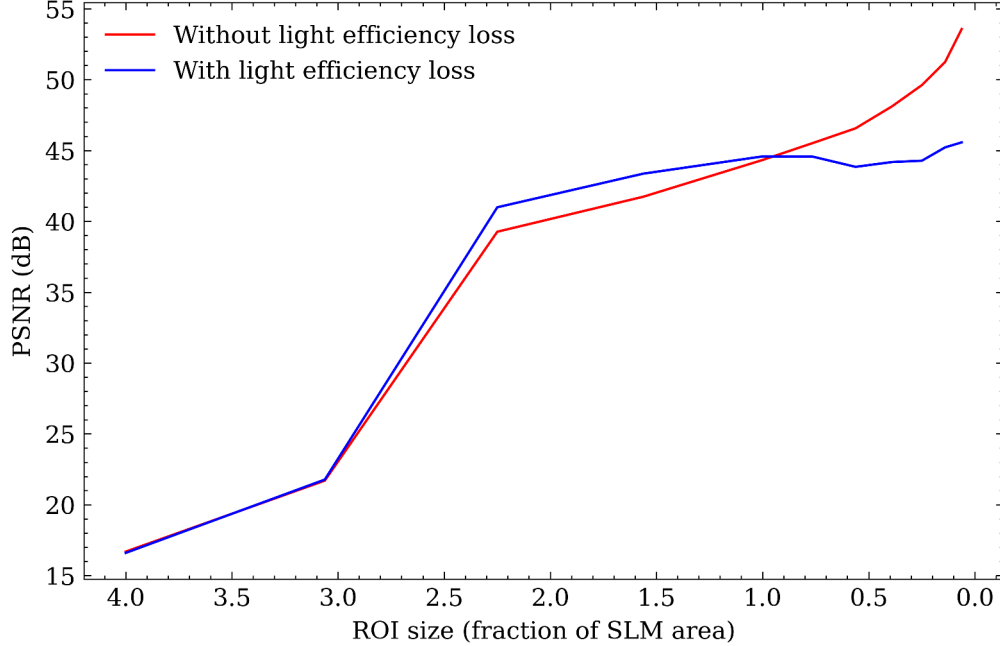


Fig. S11. Here, we optimize SLM phases to reconstruct a 2D sinusoidal pattern, as shown in Fig. S12. Optimizing a 2D hologram whose spatial resolution is larger than the SLM resolution leads to bad image reconstruction quality. As the ROI decreases (leftward on the horizontal axis), the image reconstruction quality greatly improves.

Fig. S12 shows the relationship between brightness improvement and ROI size. In our projector, the ROI size is close to the SLM size (1.0 in the plot), and we see a decent improvement in brightness. In other applications where we only need a small projection region, i.e. the ROI is small, we observe an even more drastic improvement in image brightness and light efficiency when the 2D holograms are optimized using our loss function. From the small ROI optimization results in Fig. S12(b), we see that most light is not redirected to the ROI when our loss function is not applied. The plot in Fig. S12(a) demonstrates that the brightness of a 2D sinusoidal intensity pattern improves by 10-fold for a small ROI size of 100×160 pixels when images are optimized with the light efficiency loss function. This suggests that our loss function is very crucial for small projection region applications or cases when the SLM resolution is much higher than the required projection region resolution.

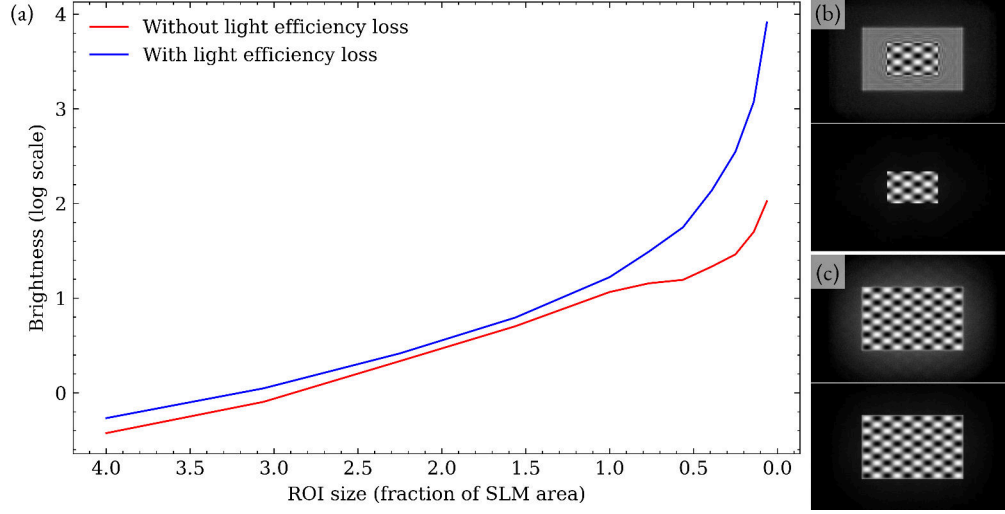


Fig. S12. (a) The brightness improvement with respect to ROI size in simulation. As the projection area decreases, the effect of our loss function becomes more significant. (b) Holograms with small ROI (1/4 of the SLM resolution) optimized without (top) and with (bottom) our loss function. The ROI is the center region with the sinusoidal pattern. (c) Holograms with large ROI (full SLM resolution) optimized without (top) and with (bottom) our loss function. Note that the image brightness here is adjusted to clearly show the light outside of the unoptimized region, and does not represent the actual image brightness.

We repeat the same experiment for a natural image and the results are shown in Fig. S13 and Fig. S14. Similar trends can be observed in both plots.

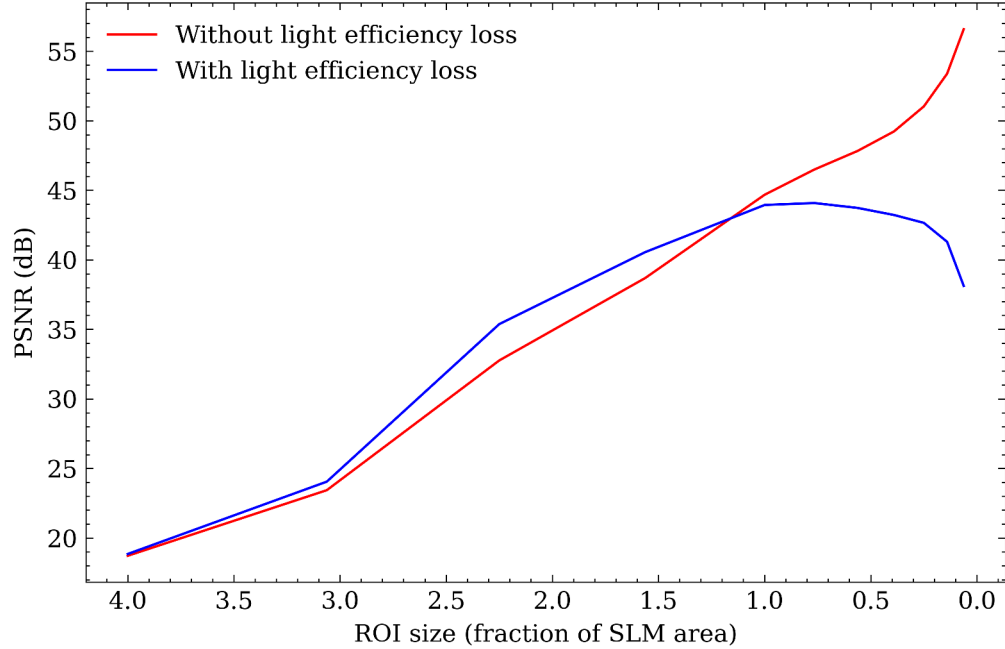


Fig. S13. The same trend in Fig. S11 can be observed for natural images. Here, we optimize SLM phases to reconstruct a 2D natural image, as shown in Fig. S14. Optimizing a 2D hologram whose spatial resolution is larger than the SLM resolution leads to bad image reconstruction quality. As the ROI decreases (leftward on the horizontal axis), the image reconstruction quality greatly improves.

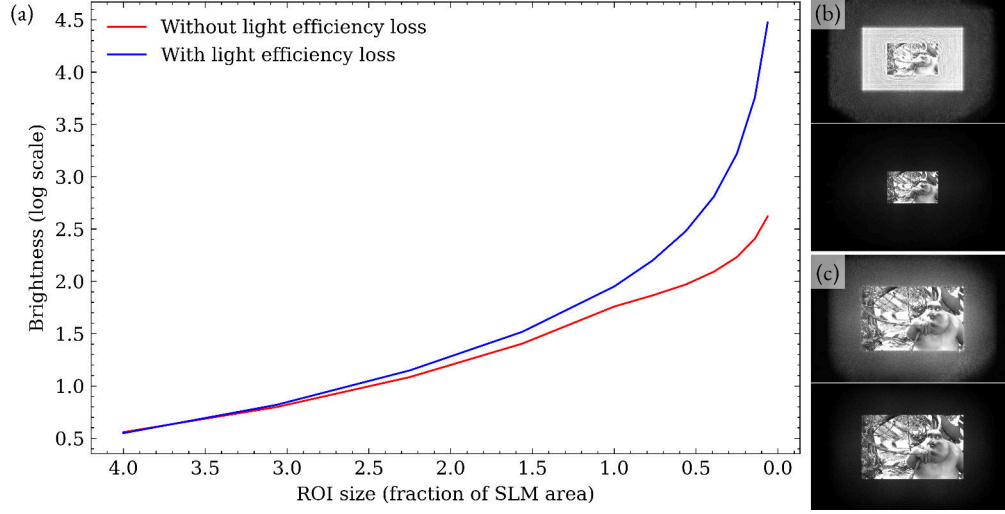


Fig. S14. The same trend in Fig. S12 can be observed for natural images. (a) The brightness improvement with respect to ROI size in simulation. As the projection area decreases, the effect of our loss function becomes more significant. (b) Holograms with small ROI (1/4 of the SLM resolution) optimized without (top) and with (bottom) our loss function. The ROI is the center region with the sinusoidal pattern. (c) Holograms with large ROI (full SLM resolution) optimized without (top) and with (bottom) our loss function. Note that the image brightness here is adjusted to clearly show the light outside of the unoptimized region, and does not represent the actual image brightness.

10. BRIGHTNESS IMPROVEMENT COMPARISON BETWEEN SIMULATION AND EXPERIMENTAL RESULTS

We average the brightness improvement of holograms when using our loss function over the whole testing set. In simulation, the brightness improvement is 66%. For the experimentally captured results, the brightness improvement is 33%. Although this seems like a fairly large difference, we believe this discrepancy can be attributed to the way we performed our brightness calculation based on the scale factor s (see Section 4 of supplementary materials). The scale factor s is computed such that the reconstructed hologram best matches the target hologram intensity in terms of pixel-wise mean-squared error. This makes the brightness calculation very sensitive to the captured image quality. In our experimental setup, there are multiple sources of error during *capture*, such as camera dark shot noise, which effects the scale factor s and hence the brightness calculation. In the future, we plan to use more precise methods to measure the total luminous flux of the projected holograms and compute actual brightness of the image.

Another source of error comes from the total energy term in Eq. 3 in the manuscript. In practice, we calculate the total allocatable energy by calculating the intensity sum of the captured image when a blank phase pattern is displayed on the SLM. However, errors such as camera dark shot noise will affect the precision of measurement. In the future, we plan to use more precise methods to measure the total luminous flux of the projected holograms and compute the actual allocatable energy of the holograms.

In conclusion, although there is a mismatch between the simulated and captured hologram brightness improvement, we believe our brightness calculation method serves as a good proxy and that more precise measurement methods can be adopted in the future to bridge the gap between experimentally captured and simulated results.

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